FIXED-PARAMETER TRACTABLE COMPUTATION OF KNOT POLYNOMIALS OF RESHETIKHIN–TURAEV TYPE

SHANA YUNSHENG LI

Abstract. We give a thorough analysis of the time complexity of computing knot polynomials of Reshetikhin–Turaev type via tensor contractions on the associated tensor networks, showing that the computation is fixed-parameter tractable with respect to a parameter that is at most linear in the treewidth of the input knot diagram. When combined with an existing algorithm for approximate tree decomposition, this gives a subexponential bound $O(e^{\sqrt{n}})$ for the time complexity of computing any knot polynomial of Reshetikhin–Turaev type, and in particular improves upon the previous bound for the HOMFLY-PT polynomial given by Burton.

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1. Introduction

Knot polynomials are one of the most important kinds of knot invariants, and have been widely studied in knot theory ever since the first knot polynomial, the Alexander polynomial [Ale28], was discovered. They are powerful tools for distinguishing knots [HTW98, Bur20, Thi25], and have deep connections with various areas of mathematics and physics. As is the case for many invariants, efficient computation is essential in providing insights for studying their properties [Shu11, GL25, HKST25], and is thus naturally an important question. Although there exist knot polynomials with intriguing properties that can be computed in polynomial time [BNvdV19, BNvdV25], the majority of knot polynomials adopt

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definitions via the Reshetikhin–Turaev functor [RT90, Tur88, GK26], and are generally $\#P$-hard to compute [JVW90], for which the best algorithm one could expect is *fixed-parameter tractable*.

Previously, fixed-parameter tractable algorithms were established for the Jones polynomial [Jon87, Mak05] and the HOMFLY-PT polynomial [FYH+85, Bur18]. Both algorithms are fixed-parameter tractable with respect to the *treewidth*, which grows at most $O(\sqrt{n})$ [LT79], where n is the number of crossings of the input knot diagram. Both the Jones polynomial and the HOMFLY-PT polynomial are of Reshetikhin–Turaev type [JM19], but those algorithms rely on specific structures of the Jones polynomial and the HOMFLY-PT polynomial, do not use the Reshetikhin–Turaev definitions, and, to the best of the author's knowledge, do not easily extend to other knot polynomials.

On the other hand, the computation of knot polynomials of Reshetikhin–Turaev type naturally translates into tensor contractions on the corresponding tensor networks, which provides a universal framework for computing these knot polynomials. Experiments with such a framework have revealed this kind of computation to be quite efficient [MK19, GL25], but no thorough analysis of its time complexity exists in previous literature. In this note, we show that such computations are in fact fixed-parameter tractable, where the parameter is in linear correspondence with the treewidth of the knot diagram. More precisely, we show the following:

Theorem 1.1. *Given an oriented knot diagram D with n crossings and a contraction sequence on D with contraction width w, for any k-variable knot polynomial RT of Reshetikhin–Turaev type, computing $RT(D)$ is*

$$O(d^w n^{2k+2}(\log n)^2), \tag{1}$$

where d is the dimension of V and V is a vector space such that the R-matrix for RT lives in $\mathrm{Aut}(V \otimes V)$.

Let $w(D)$ denote the minimal contraction width of all contraction sequences on D, and let $tw(D)$ denote the treewidth of D. The following was shown in [MS08, Section 4]:

Lemma 1.2 (Markov & Shi)**.** For any knot diagram D,

$$w(D) \leq 4(tw(D) + 1). \tag{2}$$

Combining [MS08, Proposition 4.2] and [CFK+15, Theorem 7.18], we can drop the contraction sequence from the input of Theorem 1.1, obtaining

Corollary 1.3. Given an oriented knot diagram D with n crossings, for any k-variable knot polynomial RT of Reshetikhin–Turaev type, computing $RT(D)$ is

$$O(d^{4w(D)} n^{2k+2}(\log n)^2), \tag{3}$$

where d is the dimension of V and V is a vector space such that the R-matrix for RT lives in $\mathrm{Aut}(V \otimes V)$.

Applying Lemma 1.2 and substituting $tw(D)$ with $O(\sqrt{n})$, this translates to

Corollary 1.4. Given an oriented knot diagram D with n crossings, any knot polynomial of Reshetikhin–Turaev type of D can be computed in time $e^{O(\sqrt{n})}$.

In particular, for the HOMFLY-PT polynomial, Corollary 1.4 improves upon the time complexity bound $e^{O(\sqrt{n}\log n)}$ established in [Bur18, Corollary 21].

We give a brief review of the Reshetikhin–Turaev functor in Section 2, explaining how the computation translates into tensor contractions, and then give the detailed analysis for the time complexity in Section 3, proving Theorem 1.1. More details on the material in Section 2 can be found in [GL25]. Throughout this paper, knot polynomials of Reshetikhin–Turaev type are assumed to be given by R-matrices and ribbon elements with matrix representations whose entries are Laurent polynomials with integer coefficients.

We remark that our results easily generalize to links, but we talk about knots exclusively for simplicity and because "knot polynomial" is a much more common phrase than "link polynomial".

2. Setup

We give some basic descriptions about our settings in this section, laying the groundwork for the discussion of time complexity in Section 3.

2.1. Long knot diagrams and rotation numbers.

Given a planar diagram of an oriented knot, we convert it into an oriented long knot diagram by cutting an arc open and pulling the two open ends towards infinity in opposite directions without creating any self-intersection. Following the convention in [BNvdV24], we regard the out-pointing strand to be pulled vertically upward, and the other one vertically downward. We further require that the arcs always point upward in sufficiently small neighborhoods of each crossing by applying an ambient isotopy which may rotate the crossings.

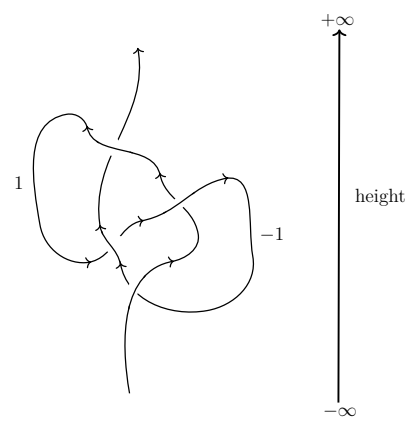


Figure 1. An oriented long knot diagram of the 4_1 knot, with nonzero rotation numbers labeled.

The vertical axis now gives us a natural height function, which restricts to a Morse function along the long knot diagram, whose local maxima and minima occur only on the arcs and never at the crossings. The rotation number associated to each arc in the oriented long knot diagram is defined as follows.

Definition 2.1. The *rotation number* of an arc $\mathcal{A}$ in an oriented long knot diagram equipped with a height function ℓ as described above, is the integer

$$\sum_{p\in\{d\ell=0\}\cap\mathcal{A}} (-1)^{\delta_\ell(p)}\varepsilon_{\mathcal{A}}(p),$$

where $\delta_\ell(p) = 1$ if p is a local maximum of ℓ and 0 otherwise; $\varepsilon_{\mathcal{A}}(p) = 1$ if the oriented arc $\mathcal{A}$ points to the right at p, and 0 otherwise.

Note that the rotation number is determined not only by the combinatorial information of the oriented long knot diagram, but also by how the diagram is embedded in the plane, inducing the height function. Given a planar diagram of an oriented knot with n crossings, there is an $O(n)$ algorithm that converts it into an oriented long knot diagram with rotation numbers in $\{0, \pm 1\}$ [BNvdV24]. A description of the algorithm can be found in [GL25, Section 2.2.1].

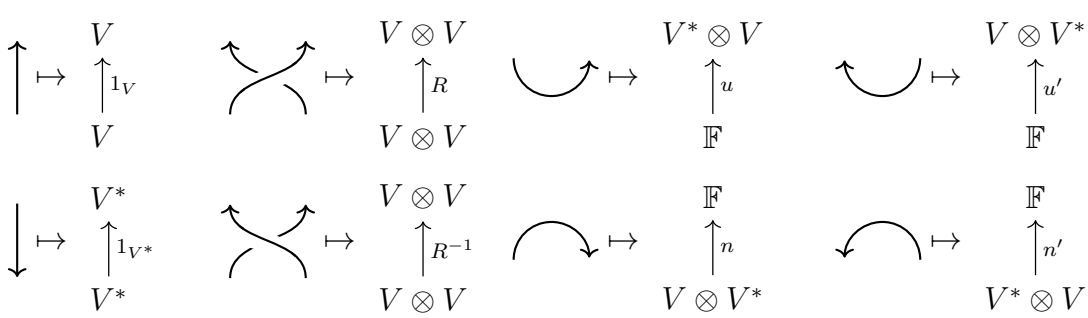
2.2. **The Reshetikhin–Turaev functor.** Depending on the context, the precise constructions of the Reshetikhin–Turaev functor vary, but they all share some common structures, of which only those relevant to computation are described here; we refer the reader to [Oht02, GK26] for in-depth introductions.

The central idea of the Reshetikhin–Turaev functor is to view oriented long knot diagrams as an analogue of 1-dimensional cobordisms, assembled from elementary pieces, and to map the pieces into the category of vector spaces. The composition of the obtained linear maps is thus the desired knot invariant.

Therefore, as with any topological quantum field theory, the Reshetikhin–Turaev functor is determined by its image on those elementary pieces. The information needed consists of:

- A vector space V over a field $\mathbb{F}$.
- An *R-matrix* $R \in \mathrm{Aut}(V \otimes V)$.
- A ribbon element $h \in \mathrm{Aut}(V)$.

The Reshetikhin–Turaev functor is then determined as follows:



where u and u' are defined by

$$u(1) = \sum_i e_i^* \otimes h(e_i), \quad u'(1) = \sum_i e_i \otimes e_i^*,$$

where $\{e_i\}$ is any basis of V and $\{e_i^*\}$ is the dual basis of $\{e_i\}$, and n and n' are defined by

$$n(x \otimes f) = f(h^{-1}(x)), \quad n'(f \otimes x) = f(x)$$

for any $x \in V$ and $f \in V^*$.

Since $(n \otimes 1_V) \circ (1_V \otimes u) = 1_V$ and $(n' \otimes 1_{V^*}) \circ (1_{V^*} \otimes u') = 1_{V^*}$, it is easy to see that the result of applying the Reshetikhin–Turaev functor is determined by the combinatorial information of the planar diagram along with the rotation numbers. With the rotation numbers living in $\{0, \pm 1\}$, the result can be described by $O(n)$ elementary pieces, hence it is $O(n)$ to convert an oriented long knot diagram into compositions of linear maps as above.

The result of applying the Reshetikhin–Turaev functor to an oriented long knot diagram is a diagonalizable linear map in $\mathrm{End}(V)$ whose eigenvalues are all the same. To obtain a knot polynomial, indeterminates are plugged into the definitions of R and h, turning the resulting linear map into a parametrized family, and the eigenvalue into the polynomial.

2.3. **Tensor network and contractions.** We briefly review some standard terminology of tensor networks and tensor contractions. We refer the reader to [GL25] for slightly more detailed descriptions, and to [BC17] for an extensive and graphical introduction to this topic.

Definition 2.2. Let $\mathcal{R}$ be a ring. An *n-tensor* T over $\mathcal{R}$ is a tuple $(T_{i_1,\dots,i_n})_{(i_1,\dots,i_n)\in\mathcal{S}}$ where $T_{i_1,\dots,i_n} \in \mathcal{R}$ and the index set $\mathcal{S}$ is of the form

$$\mathcal{S} = \{1, \dots, m_1\} \times \cdots \times \{1, \dots, m_n\} \subset \mathbb{N}^n,$$

We call the integer m_k ($k \in \{1, \dots, n\}$) the dimension of the *leg* k of the tensor T.

Definition 2.3. A *tensor network* over ring $\mathcal{R}$ is a (multi)graph Γ where each vertex v is associated with a tensor T_v over $\mathcal{R}$ whose legs are bijectively associated with the ends of the edges incident to v. Such an association must satisfy that for each edge in Γ, its two ends are associated with two different legs with the same dimension of one or two tensors.

Given a tensor network Γ over $\mathcal{R}$, for an n-tensor T and an n'-tensor T' in Γ, if two legs k of T and k' of T' are associated with the two ends of an edge e in Γ, we can *contract T and T' along e* to obtain an $(n+n'-2)$-tensor T'', defined by

$$T''_{i_1,\dots,\widehat{i_k},\dots,i_n,j_1,\dots,\widehat{j_{k'}},\dots,j_{n'}} := \sum_{i_k = j_{k'} \in \{1,\dots,m_k\}} T_{i_1,\dots,i_k,\dots,i_n} T'_{j_1,\dots,j_{k'},\dots,j_{n'}}$$

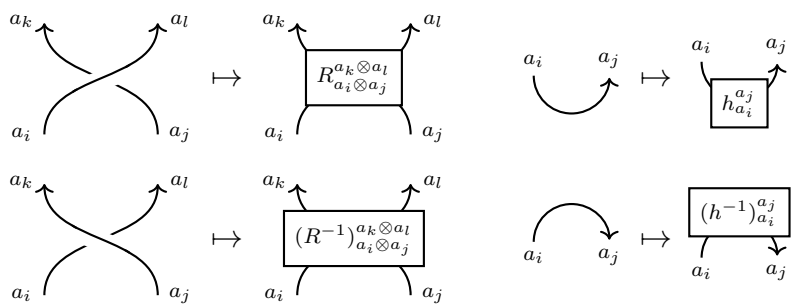
where the hats indicate that the corresponding indices are deleted. Moreover, we can remove the edge e and replace T and T' and their corresponding vertices with T'' and a new vertex, obtaining a tensor network with one fewer vertex and edge than Γ.

More generally, if legs $k_1, \dots, k_s$ of T and legs $k'_1, \dots, k'_s$ of T' are associated with the two ends of edges $e_1, \dots, e_s$ in Γ respectively, we can contract T and T' along $e_1, \dots, e_s$ at once to obtain an $(n+n'-2s)$-tensor T'' defined by

$$T''_{i_1,\dots,\widehat{i_{k_1}},\dots,\widehat{i_{k_s}},\dots,i_n,j_1,\dots,\widehat{j_{k'_1}},\dots,\widehat{j_{k'_s}},\dots,j_{n'}} := \sum_{\substack{i_{k_1} = j_{k'_1} \in \{1,\dots,m_{k_1}\} \\ \vdots \\ i_{k_s} = j_{k'_s} \in \{1,\dots,m_{k_s}\}}} T_{i_1,\dots,i_n} T'_{j_1,\dots,j_{n'}} \tag{4}$$

We can then remove the edges $e_1, \dots, e_s$ and replace T and T' and their corresponding vertices with T'' and a new vertex, obtaining a tensor network with one fewer vertex and s fewer edges than Γ. Similarly, we can define contractions of more than two tensors at once.

Example 2.4. Given an R-matrix $R \in \mathrm{Aut}(V \otimes V)$ and a ribbon element $h \in \mathrm{Aut}(V)$ determining a Reshetikhin–Turaev functor, we fix a basis $\mathcal{B} := \{e_1, \ldots, e_d\}$ of V. Under $\mathcal{B}$, $R^{\pm 1}$ becomes a 4-tensor $(R^{\pm 1})^{e_k \otimes e_l}_{e_i \otimes e_j}$ as (i, j, k, l) varies in $\{1, \ldots, d\}^4$. Similarly $h^{\pm 1}$ becomes a 2-tensor $(h^{\pm 1})^{e_j}_{e_i}$. For an oriented long knot diagram with n crossings, the Reshetikhin–Turaev functor turns it into a tensor network by the following assignment:



That is, it turns each crossing into a degree 4 vertex associated with a 4-tensor $(R^{\pm 1})^{e_k \otimes e_l}_{e_i \otimes e_j}$, and each cup or cap going rightwards into a degree 2 vertex associated with a 2-tensor $(h^{\pm 1})^{e_j}_{e_i}$. With rotation numbers in $\{0, \pm 1\}$, we obtain a tensor network with no more than $2n$ vertices, where all legs are of dimension d.

Example 2.5. Figure 2 illustrates the tensor network obtained by applying the Reshetikhin–Turaev functor on an oriented long knot diagram of the 4_1 knot. With the symbols in Figure 2, one can check that the corresponding knot polynomial, as discussed at the end of Section 2.2, can be expressed as the following finite sum:

$$\sum_{\substack{a_1, \ldots, a_9 \in \mathcal{B} \\ a_0 = a_{10} = e_1}} R^{a_8 \otimes a_1}_{a_0 \otimes a_7} \cdot h^{a_4}_{a_3} \cdot (R^{-1})^{a_9 \otimes a_5}_{a_4 \otimes a_8} \cdot R^{a_2 \otimes a_6}_{a_5 \otimes a_1} \cdot (h^{-1})^{a_7}_{a_6} \cdot (R^{-1})^{a_3 \otimes a_{10}}_{a_9 \otimes a_2}. \tag{5}$$

Recall that $\mathcal{B} = \{e_1, \ldots, e_d\}$ is the basis of V.

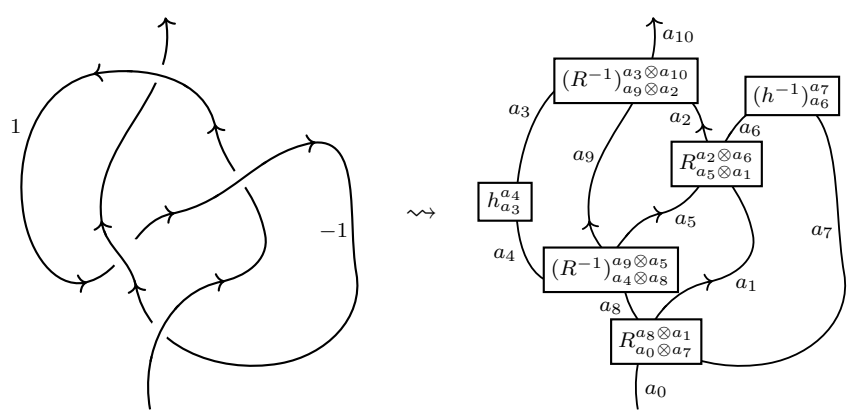


FIGURE 2. The tensor network resulting from an oriented long knot diagram of the 4_1 knot.

Examples 2.4 and 2.5 show that all knot polynomials of Reshetikhin–Turaev type can be expressed as a sum similar to (5). The sum is finite, hence we can compute it piece by piece in an arbitrary order, which naturally translates into doing a sequence of tensor contractions on the tensor network until only one vertex remains. Such a sequence is called a *contraction sequence*. A contraction sequence can be encoded by an ordered list of sets of edges removed at each step, and it plays a crucial role in the time complexity of computing the contractions, which we discuss in detail in the following section. Note that we can eliminate two free legs of the tensor network by setting $a_0 = a_{10} = e_1$ in Example 2.5 at the beginning.

3. Bounding the time complexity

3.1. **Contraction width.** We now restrict ourselves to tensor networks obtained by applying the Reshetikhin–Turaev functor to oriented long knot diagrams, where all legs of tensors have dimension $d = \dim V$.

It is clear from (4) that it takes $O(c \cdot d^s)$ additions and multiplications in $\mathcal{R}$ to compute a single entry in a tensor resulting from contracting c tensors along s edges at once. Let $l_1, \ldots, l_c$ be the number of legs of the c tensors, the resulting tensor thus has $-2s + \sum_{i=1}^{c} l_i$ legs, hence consists of $d^{-2s+\sum_{i=1}^{c} l_i}$ entries. Therefore the total number of additions and multiplications required to build the resulting tensor is

$$O(c \cdot d^s) \cdot d^{-2s+\sum_{i=1}^{c} l_i} = O(c \cdot d^{-s+\sum_{i=1}^{c} l_i}).$$

This motivates the following definition.

Definition 3.1. Given a contraction sequence $\mathcal{S}$ on a tensor network Γ, for a step in $\mathcal{S}$ that contracts c tensors with $l_1, \ldots, l_c$ legs respectively along s edges at once, its *local contraction width* is $-s + \sum_{i=1}^{c} l_i$. The *contraction width* $w(\mathcal{S})$ of $\mathcal{S}$ is the maximum local contraction width of all its steps, and the contraction width $w(\Gamma)$ of Γ is the minimal contraction width of all possible contraction sequences on it.

Note that adding degree 2 vertices does not affect the contraction width of a tensor network, and that the contraction width is independent of the dimension d. Hence it makes sense to consider (abstract) contraction sequences on a knot diagram D and define its contraction width $w(D)$ similarly, as we did in Section 1.

It is clear that adding more tensors to a step of contraction never decreases its local contraction width, hence at best has no effect on the exponential factor of the number of additions and multiplications required. Therefore we further restrict ourselves to contraction sequences where each step contracts exactly 2 tensors. In particular, for a tensor network with $2n$ vertices, all contraction sequences consist of $2n-1$ steps. For a contraction sequence $\mathcal{S}$ with width w, it then takes $O(n \cdot d^w)$ additions and multiplications in $\mathcal{R}$ to perform all steps in $\mathcal{S}$.

In [MS08], Markov and Shi restricted themselves to contraction sequences where each step contracts along only one edge. For such a contraction sequence $\mathcal{S}$, they defined its *contraction complexity* $cc(\mathcal{S})$ as the maximum degree of a merged vertex during the contraction process, and hence the contraction complexity $cc(\Gamma)$ of a tensor network Γ is the minimum complexity of all possible $\mathcal{S}$ where each step contracts along only one edge. It is immediate from definition that $w(\mathcal{S}) = cc(\mathcal{S}) + 1$, hence $w(\Gamma) \leq cc(\Gamma) + 1$, where the inequality may be strict

since the contraction width is defined over a larger family of contraction sequences. This translates [MS08, Lemma 4.4] into Lemma 1.2 as we stated.

3.2. **Proof of Theorem 1.1.** Given an oriented knot diagram D with n crossings and a contraction sequence on D with contraction width w, we cut an arbitrary arc in D open, turning it into an oriented long knot diagram D'. As discussed in Section 2.2, the arc cut open can be eliminated immediately after we apply the Reshetikhin–Turaev functor to D', obtaining a tensor network with at most $2n$ vertices along with a contraction sequence whose width is at most w. This procedure is $O(n)$ and is dominated by subsequent computation.

We have established in the previous section that it takes $O(n \cdot d^w)$ additions and multiplications in $\mathcal{R}$ to compute the desired knot polynomial, hence it remains only to bound the time complexity of performing additions and multiplications in $\mathcal{R}$.

For a k-variable knot polynomial of Reshetikhin–Turaev type, we have $\mathcal{R} = \mathbb{Z}[t_1^{\pm 1}, \dots, t_k^{\pm 1}]$. Let M be the maximum degree of the entries of $(R^{e_k \otimes e_l}_{e_i \otimes e_j})$ and $(h^{e_j}_{e_i})$. The maximum degree of polynomials that can appear throughout the process is thus $2nM$. Hence all polynomials have at most $\binom{2nM+k}{k} \leq (2nkM)^k = O(n^k)$ terms. Therefore the number of additions and multiplications in $\mathbb{Z}$ required for each polynomial addition or multiplication is $O(n^{2k})$.

We now bound the integers that may appear during the process. Let C be the maximum absolute value of coefficients of the entries of $(R^{e_k \otimes e_l}_{e_i \otimes e_j})$ and $(h^{e_j}_{e_i})$. Since there are at most $\binom{M+k}{k} \leq (kM)^k =: A$ distinct monomials appearing in the entries, the maximum absolute value of integers that can appear throughout the process is bounded by the maximum absolute value of the coefficients of

$$\left(\sum_{i=1}^{A} C m_i \right)^{2n} = \sum_{\substack{p_1 + \cdots + p_A = 2n \\ p_1, \dots, p_A \geq 0}} \frac{(2n)! \cdot C^{2n}}{(p_1!) \cdots (p_A!)} \cdot m_1^{p_1} \cdots m_A^{p_A}. \tag{6}$$

where each m_i represents a monic monomial. Hence it is bounded by

$$\sum_{\substack{p_1 + \cdots + p_A = 2n \\ p_1, \dots, p_A \geq 0}} \frac{(2n)! \cdot C^{2n}}{(p_1!) \cdots (p_A!)} \leq \binom{2n + A}{A} \cdot (2nC)^{2n} \leq (2nCA)^{2n+A}, \tag{7}$$

so the bit-size of the integers is $O(n \log n)$. Therefore the result follows from the fact that adding and multiplying integers with bit-size in $O(n \log n)$ are

$$O((n \log n) \cdot \log(n \log n)) = O(n (\log n)^2). \tag{8}$$

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(Shana Yunsheng Li) Department of Mathematics, University of Illinois, Urbana, IL, USA
https://shana-y-li.github.io

Email address: yl202@illinois.edu